

Chapter 5: Production

Meaning of Production

Production refers to the **transformation of inputs (resources or factors of production)** into **outputs (goods and services)**.

Example: Inputs like **sugarcane (raw material)**, **capital (machines)**, and **labour (workers)** are used to produce **sugar**.

Production Function

A **production function** shows the **technical relationship** between **inputs (factors of production)** and **output**, under a given technology.

$$Q = f(L, K)$$

where,

- Q = Quantity of output
- L = Labour
- K = Capital

It shows **how much output** can be produced from **different combinations of labour and capital**.

Types of Production Function

1. Short-run Production Function

In the short run, the **law of variable proportions** studies the **input–output relationship** where **only one factor of production is variable** while others remain fixed.

$$Q = f(L, \bar{K})$$

Where labor (L) is variable factor and capital ($\bar{K}$) is fixed factor.

2. Long-run Production Function

The input and output relationship in the long run is studied under the law of return to scale. It is also known as production function with two variable input.

$$Q = f(L, K)$$

Where labor (L) and capital (K) are variable factor.

a. Total Product (TP)

Total Product refers to the **total quantity of goods produced** using a given quantity of inputs in a specific period.

$$TP = \sum MP$$

or

$$TP = AP \times L$$

- It increases as more of the variable factor (like labour) is added — but at different rates.

b. Average Product (AP)

Average Product is the **output per unit of the variable factor**.

$$AP = \frac{TP}{L}$$

c. Marginal Product (MP)

Marginal Product is the **additional output produced by employing one more unit of the variable factor**.

$$MP = TP_n - TP_{n-1}$$

or

$$MP = \frac{\Delta TP}{\Delta L}$$

Relationship between MP and AP

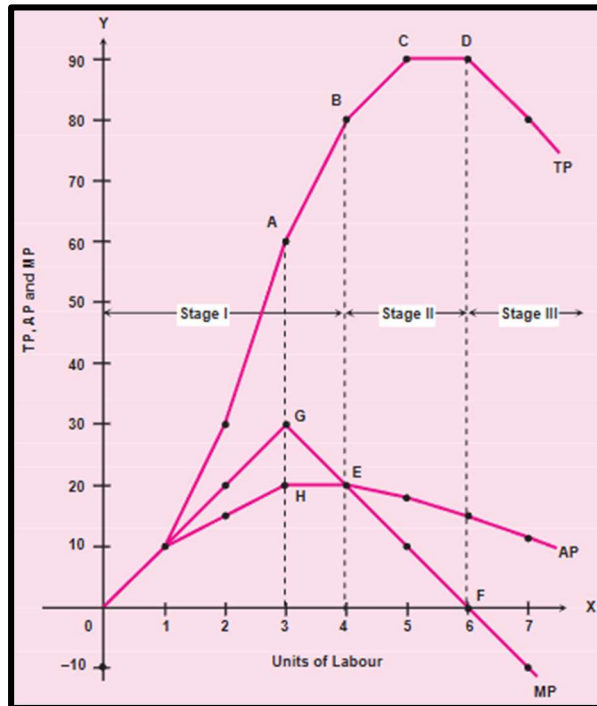


- When $MP > AP$, AP is rising.
- When $MP = AP$, AP is at its maximum.
- When $MP < AP$, AP is falling.

- iv. MP may be positive, negative, or zero but AP is always positive.
- V. When MP is rising, AP must rise but when MP is falling, AP may rise or fall.

4. Law of Variable Proportions/Law of Diminishing returns

The **Law of Variable Proportions** explains how output changes when **one input is varied**, while **others are kept constant**.



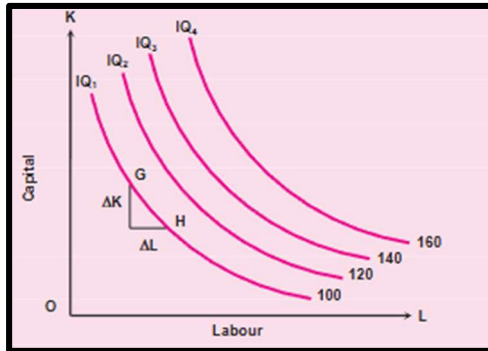
- It applies to the **short run**, where at least one factor (like capital) is **fixed**.
- As more units of the variable factor (like labour) are added, output passes through **three stages**:

Stages of Production:

Stage	Name	Behavior of TP, AP, MP	Cause
I	Increasing Returns	TP ↑ at increasing rate, MP ↑	Efficient use of fixed and variable factors
II	Diminishing Returns	TP ↑ at decreasing rate, MP ↓ but positive	Optimum stage (rational producer operates here) where scarcity of fixed factor occurs
III	Negative Returns	TP ↓, MP becomes negative	Overuse of variable factor

Rational Stage of Production: Stage II (Diminishing Returns)

5. Isoquant and Isoquant Map



- **Isoquant:** Curve showing **different combinations of two inputs (L and K)** that produce the **same level of output**.
 - **Isoquant Map:** A set of isoquants representing **different levels of output** (similar to contour lines on a map).
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6. Returns to Scale

Returns to scale describe how output changes when **all inputs are changed in the same proportion**.

(It is a **long-run concept**.)

Types:

1. **Increasing Returns to Scale:**
Output increases **more than proportionately** to inputs.
e.g., Inputs ↑ by 50%, Output ↑ by 80%
2. **Constant Returns to Scale:**
Output increases **in the same proportion** as inputs.
e.g., Inputs ↑ by 50%, Output ↑ by 50%
3. **Decreasing Returns to Scale:**
Output increases **less than proportionately** to inputs.
e.g., Inputs ↑ by 50%, Output ↑ by 30%